

Behavioral Study Analysis

How Interruptions of Control in High and Low Stakes Situations Affect Attention

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Abstract

Interruptions of control seem to result in narrowed focus, especially in high stakes situations. Previous work (see Wen & Haggard, 2018) has been done exploring control and attention, but not with stakes nor with consideration of attention to task-noncritical stimuli. The nature of attention in high-stakes situations is important for understanding where people are focusing in critical contexts, such as hydroplaning while driving. We hypothesized that participants playing a web game would take longer to react to noncritical stimuli on the edge of the screen, especially in high-stakes situations or when they lost some control, compared to similar stimuli in the middle. We also hypothesized that the disparity (between central and peripheral) in reaction times would get even larger when both high stakes and loss of control happened at the same time. Results showed that participants' reaction times do get worse in either individual condition (loss of control or high stakes situations) but not interactively; in high stakes situations with lost control, reaction time was similar to either condition individually (high stakes *without* control lost or low stakes *with* control lost). The results, though limited, offer support for models from other literature. We finally outline a call for future research using a technique for multiplying the effect discovery capabilities of a single study.

Introduction

The relationship between control and agency is a widely debated topic. While there is a range of subjective experiences relating to the subject, it has proved especially difficult to study in an experimental context. In a 2018 study, researchers Wen and Haggard tackled this idea by examining how the gain or loss of control may affect one's attention. Their study, *Control Changes The Way We Look at the World* (Wen & Haggard, 2018), references the idea of error

monitoring, a concept outlined by Blakemore et al. (2001). Foundational to Blakemore et al.'s idea of error monitoring is the comparator model, which seeks to explain how an individual may neurologically respond to unpredicted sensory stimuli. This model relies upon the cognitive comparison between predicted sensory feedback and external influences on the sensorimotor system. From this idea, Wen and Haggard's 2018 study experimented with how interruptions of control may affect one's sense of agency and attention.

Wen and Haggard's concept draws attention to experiences of losing control in everyday life, especially when the consequences of losing control are higher. One example of this experience is losing control of a vehicle on the road, which is often marked by the feeling of one's focus narrowing rapidly on the crisis at hand. Our research examines this sensation more deeply; in particular, if a loss of control causes one's perception to *literally* narrow to a more central point. Further interesting is the element of "stakes," or the severity of consequences for a loss of control. For instance, when traveling at high speeds on the road, suddenly hydroplaning can result in life-threatening consequences if attention is not properly directed. Subjectively, these more serious situations feel as though they affect our perception more than a loss of control in a less-critical context would. On the other hand, lower-stakes incidents (like sliding on ice while driving slowly through a neighborhood) may feel as though they don't affect our attention as strongly.

Based on the subjective qualia of these experiences, we suspected that these factors may compound upon each other in moments of crisis. The question we intended to answer was whether an increase of stakes during a loss of control results in a narrowing of perception beyond the narrowing in a similar low-stakes situation. If this effect is real and these factors are indeed

connected, it may serve as a unique insight into how control loss in high-stress situations can substantially change our perception.

Literature that addresses the way loss of control in higher stakes contexts may affect perception is extremely limited, though research exploring how stakes may affect cognitive behavior and control offers interesting insight into the ways in which people respond under different levels of pressure. For example, in a Parke et al. (2015) study on how stakes affect cognitive control in gambling, it was found that higher stakes gambling scenarios had a negative effect on participants' decision making, causing them to perform more poorly in a simple probability task. Additionally, Ariely et al. (2009) questioned how incentives affect performance, finding that extremely high incentives cannot be relied upon to improve performance, suggesting that the Yerkes-Dodson law (moderate stress can enhance performance, but excessive stress and low stress can hinder performance) applies in the context of high and low stakes. While these findings are not directly related to our field of interest, they do show that an increase in stakes can have varied effects on an individual's cognition.

Methods

All stimuli, experiment scripts, data, and analysis scripts are available on the Open Science Framework at <https://osf.io/kc9gd/>. The study pre-registration is available at <https://osf.io/vnh7x>.

Participants

The study was posted on Prolific for participants to play remotely. The game was titled “Play A Video Game To Help Mr. Frog Get Bugs.” Participants with a screen size less than 1400px x 900px in either dimension were not able to participate because screen sizes smaller than this resulted in graphical issues with the game, as revealed by testing. Before beginning the game, players were asked only for basic demographic information: age, gender identity, dominant hand (right or left), hours of gameplay per week, and device (laptop or desktop). To incentivize effort from the participants, they were told that if they scored the highest, they would be paid a \$6 bonus. We aimed to recruit 97 participants, calculated by dividing our total budget after the \$6 incentive (\$194) by the payment per participant (\$2). Due to a miscommunication, we ended up with 64 participants since each participant was paid \$3 instead of \$2. This was an unintentional deviation from our pre-registration.

Procedure

Participants were given instructions on how to play the game along with a confirmation quiz to ensure their understanding. They needed to correctly answer critical questions about the gameplay in order to continue. This was done to ensure participants understood the rules before beginning. The participants controlled a frog in an endless runner platform game, seen in Figure 1 below. To move the frog, players pressed the right or left arrow keys. There were multiple

incentives for the player. Collecting regular bugs would add 10 points to their score. Bonus bugs were an additional incentive, as well as a way to measure attention with reaction time. To collect bonus bugs, participants pressed space. Bonus bugs would appear in the center of the screen or on the periphery (left side of the screen), and collecting them would add 50 points. They appeared 800 ms after the start of the trial. Each bonus bug started at a smaller size and grew 15% of its original size every 100 ms. After 3500 ms, it would disappear from the screen if not already collected by the participant. Bonus bugs were not placed on the right side of the screen because then peripheral bonus bugs would have more possible locations than central bonus bugs, and this could introduce a difference beyond just the peripheral/central divide we were interested in.

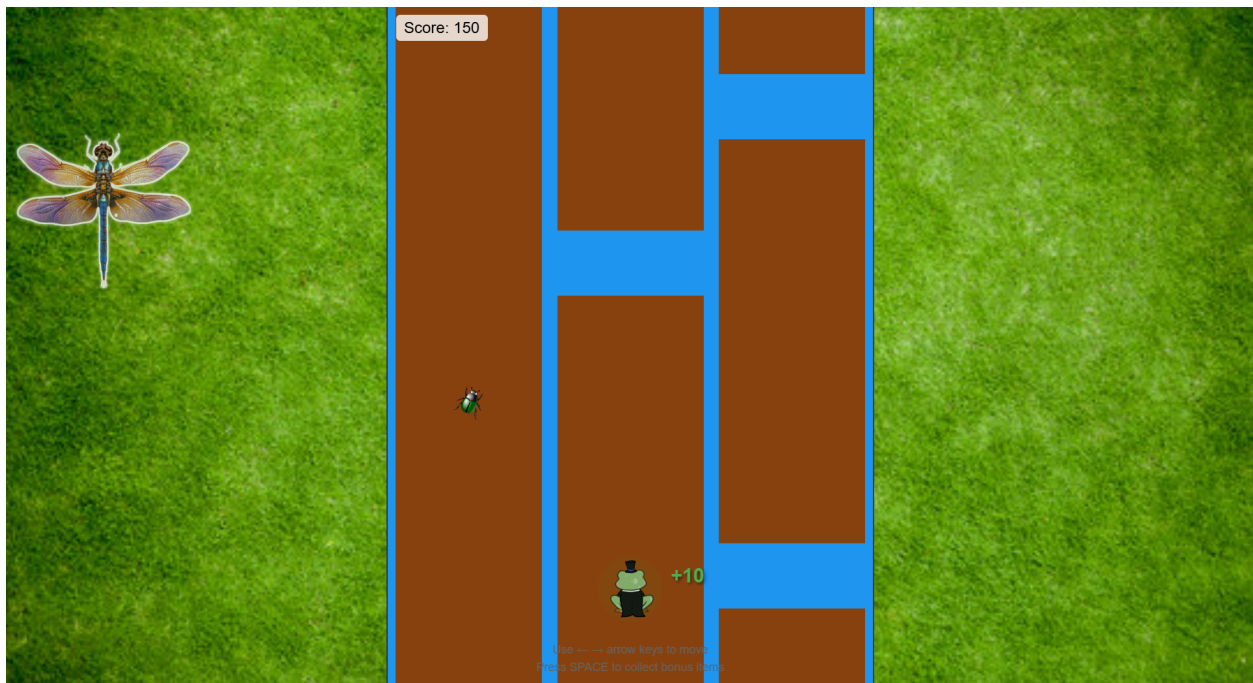


Figure 1: Screenshot of the final game (as published to Prolific)

Note. This is from the practice and does not represent any specific condition

The main objective was to collect as many points as possible without falling into the water. All participants started with a score of 10,000. Falling into the water resulted in a penalty of -100 points. To avoid false-positive bonus bug collection attempts, they were worth -50 points. Randomly throughout the game, the frog would “slip,” causing the participant’s arrow key presses to be ignored for one second. These slips were queued by the code and only executed at the next participant button press, meaning there was no advance notice of a slip; only after attempting to move would a participant realize they had lost control of their character. After completion of the confirmation quiz, participants were shown an example of what the bonus bug would look like when it appeared in the center and in the periphery of the screen. Participants were then prompted to press the spacebar when they felt ready to begin the practice round. The practice round lasted for one minute. At the end of the practice, participants were prompted again to press space to begin the trials. The trials lasted approximately 10 minutes. Trials were seamless, meaning the participants would see not discrete blocks but rather a continuous stream of obstacles and incentives.

The game had a total of eight conditions and four filler conditions. Filler conditions consisted of interrupted and uninterrupted movies as well as central or peripheral bonus bugs. The four filler conditions were not used for participant data and were there for smoother continuity of the game. Each of the main conditions was repeated ten times total but in a random order with randomly interspersed filler trials. Non-filler trials were either high stakes or low stakes. In a high stakes situation, a participants were on track (unless they moved) to hit a water obstacle, costing them 100 points. In a low stakes situation, participants were not on track with an obstacle, but a bug was available in another lane. Failure to move here only lost them the

opportunity for 10 points, a smaller inconvenience than a 100-point penalty. The conditions were as follows:

- High Stakes/Interruption/Central Bonus Bug
- High Stakes/Interruption/Periphery Bonus Bug
- High Stakes/No Interruption/Central Bonus Bug
- High Stakes/No Interruption/Periphery Bonus Bug
- Low Stakes/Interruption/Central Bonus Bug
- Low Stakes/Interruption/Periphery Bonus Bug
- Low Stakes/No Interruption/Central Bonus Bug
- Low Stakes/No Interruption/Periphery Bonus Bug

After participants completed all of the trials, they were prompted to rate and comment on their game experience as well as provide information on technical difficulties. These questions were added to determine if any replicable issues could be detected and were not used for statistical analysis. Participants started with a score of 10,000. Those who ended with a negative score were excluded (6 total) to disinclude participants with extremely poor performance, because in testing, almost everyone was able to maintain a positive score, meaning a 0-point minimum was a low threshold to achieve. Rounds where the participant didn't move were excluded. If the participant collected the bonus bug in a round where they were supposed to be interrupted but had not yet moved, the round was not counted. Rounds where the bonus bug was not collected were excluded because no reaction time could be assigned. We used data from 53 participants in total.

Results

Effect	DFn	DFd	F	p	p<.05	ges
stakes	1	41	18.28272488	0.0001109259 *		0.01320633802
bug_location	1	41	570.1280175	1.13E-25 *		0.4712203867
was_interrupted	1	41	5.199503035	0.02786275316 *		0.007804835365
stakes:bug_location	1	41	10.32749113	0.002554281114 *		0.007588826214
stakes:was_interrupted	1	41	8.671004034	0.005306168159 *		0.00390262083
bug_location:was_interrupted	1	41	5.328366985	0.02609994799 *		0.006936070915
stakes:bug_location:was_interrupted	1	41	0.2851587931	0.596223415		0.0001239701344

Figure 2: 3-way, within-participants ANOVA of bug location, stakes, and interruption of control on mean reaction time.

Main Effects

We expected to find that when the bonus bug spawned on the side of the screen, participants would take longer to respond to it. We also expected if the participants' control was interrupted right before a bonus bug appeared on screen, they would also take longer to respond to that. And if the participant was in a “high stakes” situation right before the bonus bug appeared, i.e. a situation where they would fall into the water if they didn't move, they would also take longer to respond.

We found that each of these conditions alone were statistically significant (**Figure 2**). However, the effect of interruption of control and “stakes” were negligible ($ges < .02$). The location of the bug had a big effect on participants' reaction times ($ges = .47$). We expected that bug location would have a big effect on reaction time, and that stakes would have a small effect

on reaction time. However, we expected that disrupting the participants' control over their avatar would shake them up a bit more.

Two-Way Interaction

Regardless of whether the participant was in a high stakes or low stakes situation, or if the participant's control was interrupted before they saw the bug or not, we expected that participants would take longer to react to a bonus bug on the periphery than one in the center. This is our main effect for bug location. However, *if* the participant was in the high stakes situation, we also predicted that they would take *even longer* to respond to a bug on the periphery than if they were in the low stakes situation (stakes : bug location). And *if* the participant's control was interrupted right before they saw the bug, we also predicted that they would take *even longer* to respond to a bug on the periphery than if their control had not been interrupted (interruption of control : bug location). Both of these effects were statistically significant (**Figure 2**). However, both the interaction between bug location and interruption of control, and the interaction between bug location and stakes had a negligible effect on mean participant reaction time—not what we expected ($ges < .01$).

Regardless of whether the bonus bug was in the center or on the periphery, or if they were in a high or low stakes situation, we expected that participants would take longer to respond to the bonus bug if their control over their avatar was interrupted just before the bug appeared. This is our main effect for interruption of control. However, *if* the participant was in a high stakes situation, we predicted that they would take *even longer* to respond to the bonus bug when their control was interrupted (stakes : interruption of control). This effect was statistically significant in our experiment (**Figure 2**). Despite that, the interaction between stakes and interruption of

control had a negligible effect on mean participant reaction time—not what we expected ($ges < .01$).

Three-Way Interaction

We also hypothesized that if the participant was both in a high stakes situation *and* their control over their avatar was interrupted right before the bonus bug appeared, they would take even longer to respond to a bonus bug on the periphery than one in the center of their screen, than if they were in both a low stakes situation *and* their control was not interrupted right before the bonus bug appeared. We thought there was a unique attention-narrowing effect that happened when *both* variables were introduced at the same time. If the participant lost control in a high stakes situation, they would usually fall into the water. We thought that if their loss of control led to them falling into the water, this would force them to focus on avoiding the water, even at the expense of missing the bonus bugs.

The effect of the three-way interaction on reaction time did not prove statistically significant ($p=.6$; **Figure 2**). We believed that the three-way interaction between bug location, interruption of control, and stakes had an effect on reaction time to the bonus bug, but we didn't have a strong theory about what the interaction would say about cognition or any ties to existing literature. It would have been useful to come up with a theory first, before we decided that we expected to see this effect.

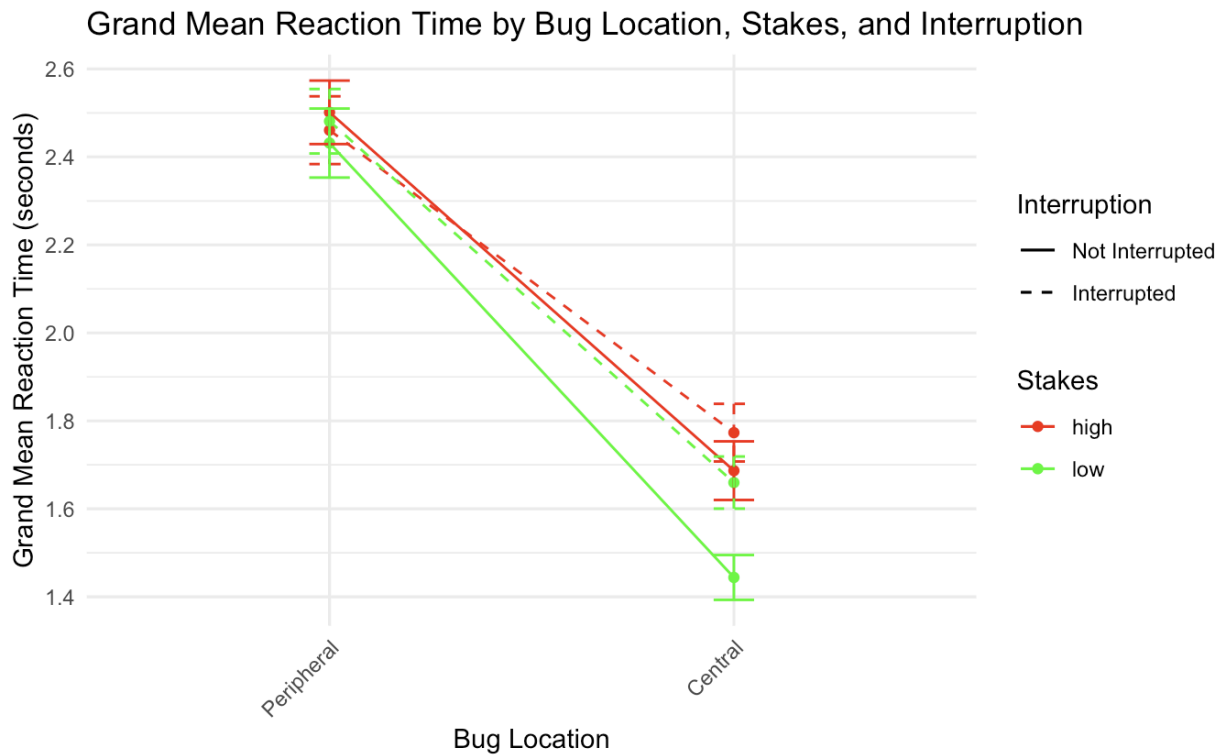
Group-Level Analysis

Figure 3: Participants responded slower to bonus bugs on the left than they did to those in the center of their screen. When the bonus bug was in the center, participants also responded faster when there were low stakes, and their control was not interrupted.

Grand mean reaction time to the bonus bugs was higher when the bug was on the side of the screen than it was when the bug was in the center of the screen. We also noticed that when the bonus bug was in the center of the screen, participant's appeared to respond to it faster when they were in the low stakes *and* uninterrupted control conditions combined (**Figure 3**).

Exploratory Analyses

In our confirmatory analysis, we only used 42 of 53 participants. Some of those participants had valid data in all but one category. didn't have enough data in only one condition. It's likely that because of the low number of trials for each condition, some participants may not have had usable data for one condition by chance. We wanted to introduce these excluded participants to our analyses by conducting three exploratory analyses, each one averaging over either bug location, interruption of control, or stakes. This was to see if excluding those participants was radically changing our results.

<u>No Control</u>						
Effect	DFn	DFd	F	p	p<.05	ges
bug_location	1	47	367.1213338	7.59E-24	*	0.4259580351
was_interrupted	1	47	5.929564178	0.01873825581	*	0.01242588543
bug_location:was_interrupted	1	47	2.10251106	0.1536983076		0.003612060367
<u>No Stakes</u>						
Effect	DFn	DFd	F	p	p<.05	ges
bug_location	1	47	367.1213338	7.59E-24	*	0.4259580351
was_interrupted	1	47	5.929564178	0.01873825581	*	0.01242588543
bug_location:was_interrupted	1	47	2.10251106	0.1536983076		0.003612060367
<u>No Location</u>						
Effect	DFn	DFd	F	p	p<.05	ges
stakes	1	51	36.64290518	1.69E-07	*	0.02393662566
was_interrupted	1	51	17.96372592	9.45E-05	*	0.02866348359
stakes:was_interrupted	1	51	21.24152417	2.74E-05	*	0.01555937538

Figure 4: Three two-way ANOVAs, each leaving out one condition in our experiment.

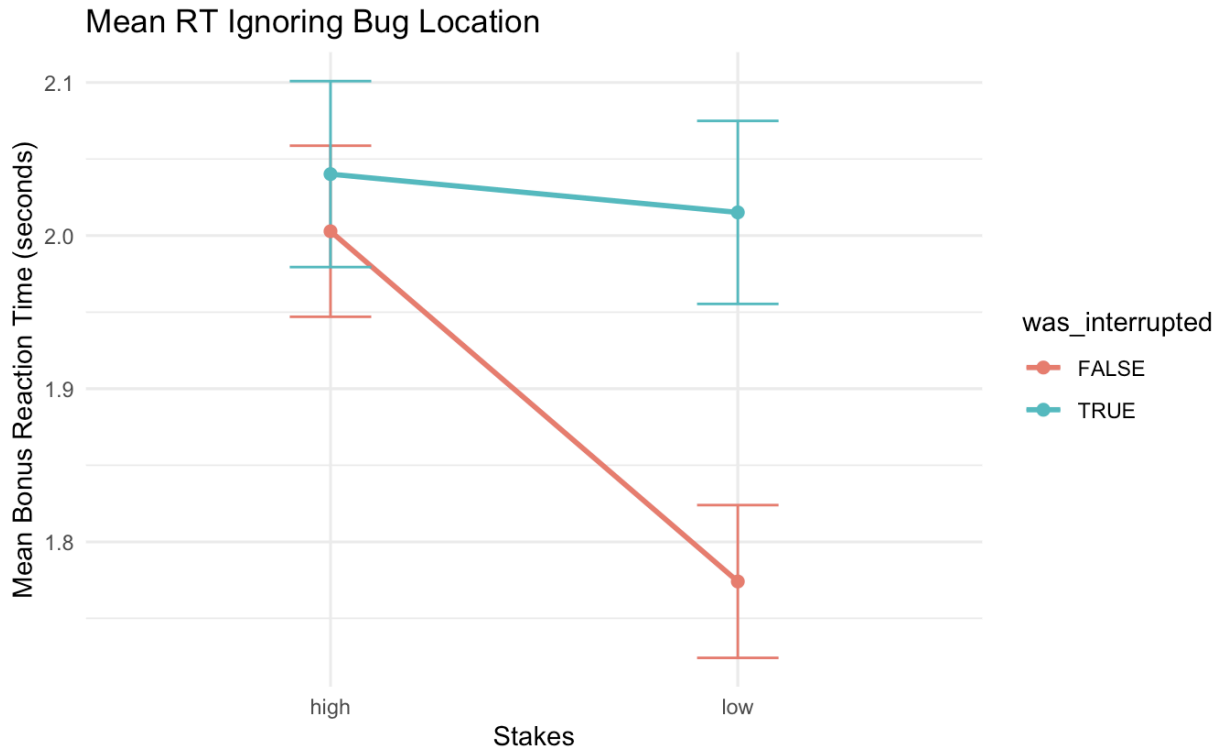


Figure 5: Participants would take longer to respond to the bonus bug if either control was interrupted or stakes were high. Participants would respond to the bonus bug faster when control was uninterrupted and stakes were low.

We were most interested in averaging over bug location in our analysis. We found that in our confirmatory analysis, bug location had a big effect on reaction time and we were concerned that this may have overshadowed the effect of stakes and interruption of control, so we conducted an ANOVA averaging over bug location with 52 participants.

We found significant results for our two main effects and our interaction effect (stakes, interruption of control, interruption of control : stakes; **Figure 4**), but again, all of the effects were small ($ges < 0.03$). As shown in **Figure 5**, we see similar reaction times for all conditions except for when stakes were low *and* control was uninterrupted, when ignoring bug location. From this, it seems that if the participant was in a high stakes situation *or* their control was

interrupted right before the bonus bug appeared, they would take longer to respond to it, but there was nothing unique about the participant being in a high stakes situation *and* having their control interrupted.

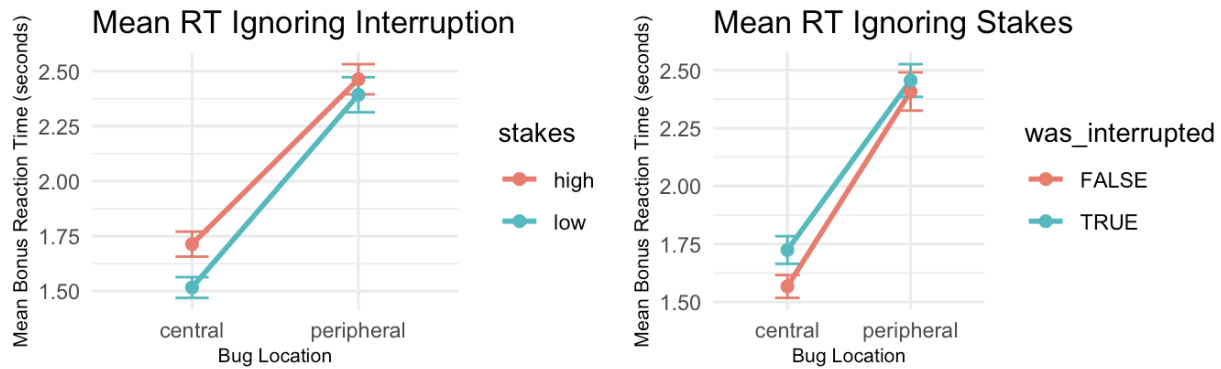


Figure 6: The relationship between reaction time and bug location is still strong when ignoring interruption of control or when ignoring stakes. The relationship between reaction time and either stakes or interruption of control is still weak, even when ignoring the other independent variable in the study (interruption of control or stakes).

We also conducted two more ANOVA tests with 48 participants each, one ignoring whether control was interrupted, and another whether stakes were high or low. We found two significant main effects for each ANOVA, but did not find significant interaction terms for either (No stakes $\rightarrow$ interruption of control, bug location, interruption of control : bug location; No interruption of control $\rightarrow$ stakes, bug location, stakes : bug location; **Figure 4**). Not finding the significant interactions is strange, and likely caused by the additional participants we used in these ANOVAs. This suggests that the two-way interactions we found between bug location and stakes, and bug location and control, are not robust.

Bug location still showed strong effects on reaction time (No stakes $\rightarrow ges = .42$; No control $\rightarrow ges = .46$) and all other effects were negligible ($ges < 0.03$). **Figure 6** shows effects of

bug location on reaction time similar to those found in our confirmatory analyses. They also show a relative lack of effect of stakes and interruption of control on reaction time, like our confirmatory analysis, suggesting that the 6 participants we added back in these analyses had little impact on our main effects.

Discussion

The lack of a three-way interaction means that the original hypothesis that reaction times would get worse interactively was wrong. The results shown in Figure 3 demonstrate that reaction times to noncritical stimuli get worse only to a point: in high stakes conditions or when control was lost, but not any more when both conditions were true.

There is scant literature that draws upon themes of attention, stakes, and control at once. Some work has been done in related but different contexts, such as in virtual reality (see Jeunet et al., 2018). Tangential research, though, can be helpful in understanding results in context to see if they align with existing proposed models and results. The study by Parke et al. (2015), which demonstrated worse performance from gamblers in higher-stakes situations shows poor performance as stakes increase. The claims of worse decision-making in their study can thus be broadened by our results about worse (for noncritical stimuli) attention. Ariely et al.'s (2009) study showed that performance gets better with moderate stakes but then worse in extreme-stakes situations. Our study only had two stake situations and thus can't confirm if participants scored better in the low-stakes situation than they would have in a no-stakes (e.g., no obstacles or rewards beyond the bonus bug) situation, but participants did perform worse in the high-stakes context than the low-stakes one. Our study could be rerun with a dedicated no-stakes condition to allow us to test whether their result holds in this context, but this development

would only be one step of progress. Almaatouq et al., (2022) push back against this sort of singly-iterative approach, which they call “twenty questions with nature” and instead argue for integrative experiment design, a novel approach which involves determining the types of variables at play in a whole category of experiments or questions and then running targeted experiments in that design space. This approach is especially applicable to online experiments given their flexibility in deployment with different parameters, even in a manner that can be automated. An expanded version of this experiment would not capture an entire experimental design space for questions about attention, control, and stakes together, given that the nature of the task as an online game already removes possible dimensions for task nature, setting, and more. However, an expanded version of the game could occupy, if not the entire design space, some subspace thereof. A group with a larger budget might consider an expanded version of our experiment with possible automated variance over many factors. Proposed dimensions of this subspace are the speed of the game, to see if faster-decision making affected certain conditions more than others; the nature of the loss of control, to examine whether delays or interruptions affect players differently; a continuous (as opposed to our discrete) bonus bug placement, to determine whether the effect on reaction time as the bonus bug gets farther is linear or not; the point values assigned to different obstacles (negative) or bonuses (positive), to see how players determine when it’s worth taking a risk despite a possible loss of control; and the nature of the reward incentive, to see if more money on the line results in worse performance, as in Ariely et al. (2009). Even more factors beyond these could be proposed and examined, but this list provides a reasonable initial set. A single experiment conducted with [Dallinger](#) or a similar tool would allow for dynamic exploration of specific points in the subspace of the above dimensions.

Given the limited research on the combination of factors in this study, it would be hasty to propose a theory or model from our results. Our results leave a question as to why high stakes and loss of control did not cumulatively affect reaction time. Hopefully, a wider-scale, dimensionalized version of this study could be used to develop a more rigorous model in which our results could contribute.

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Author Contributions

- All - study design discussions, initial planning, everyone wrote ≥ 1 sections of the draft and helped revise the draft as a whole
- Isaac - Programmed the game, wrote most of analysis code, drafted abstract and discussion
- Jackson - Preregistration, wrote some analysis code, drafted results section
- Beatrice - Collected information for prolific posting, drafted introduction section
- Irene - Made some images for the game, experiment rough design, preregistration, drafted methods section